Substitution method

1

a $x = -8, y = 8$

b $x = -5, y = 3$

c $x = -8, y = -3$

d $x = -6, y = -3$

e $x = -3, y = 5$

f $x = 6, y = 4$

g $x = 9, y = 5$

h $x = -2, y = 9$

i $x = 1, y = 6$

j $x = 8, y = 4$

k $x = -5, y = -8$

l $x = 8, y = 1$

m $x = -\frac{4}{15}, y = \frac{13}{3}$

n $x = 7, y = 4$

2

a $x = \frac{4 + 7y}{3}$

b Infinitely many solutions.

c Dependent

3

a $y = -\frac{64}{3} - \frac{5x}{3}$

b $x = -8$

c $y = -8$

4

a $x = 7, y = 4$

b $x = -6, y = 5$

c $x = 6, y = -5$

d $x = 2, y = 3$

e $f = \frac{2}{5}, g = \frac{6}{5}$

f $q = \frac{4}{5}, p = \frac{1}{8}$

g $a = \frac{3}{2}, b = \frac{5}{2}$

h $v = \frac{2}{5}, t = \frac{5}{4}$

5 By substituting the ordered pair into each equation and seeing if it satisfies them.

6 Yes, it is a solution.

7 Yes, it is a solution.

8

a $x = 4, y = 2$

b Yes

c Yes

Applications

9

a $x = y + 10$

b $x + y = 28$

c $y = 9$

d $x = 19$

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10

a $y = x + 6$ **b** $x + y = 22$ **c** $x = 8$ **d** $y = 14$

11

a $y = 10x, y = 7x + 12$ **b** $x = 4$

c $y = 40$

12

a $y = 5x, y = 9x - 32$ **b** $x = 8$

c $y = 40$

13

a $x + y = 3000, 0.06x + 0.08y = 204$ **b** $x = \$1800$ and $y = \$1200$

14

a $2f + 3g = 10.08$ and $5f + 7g = \$24.08$ **b** $f = \$1.68$ and $g = \$2.24$

15

a $y = 4x - 190\,000,\ y = x + 560\,000$ **b** $x = 250\,000$ and $y = 810\,000$

16

a $x + 10y = 105$ and $x + 15y = 155$ **b** $x = 5,\ y = 10$

c $\$255$

17 Each racket is $\$500$ and each ball is $\$50$.

18 15 pieces of $10c$ coins and 45 pieces of $5c$ coins

19 35 hotdogs and 52 sodas.

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